Group 4 – Normal Distribution

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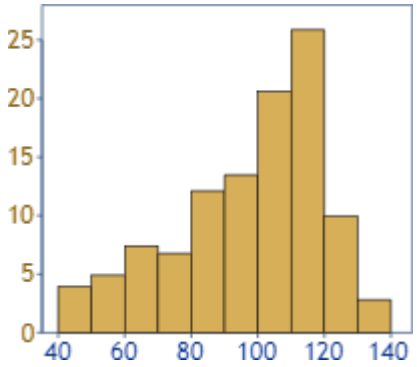
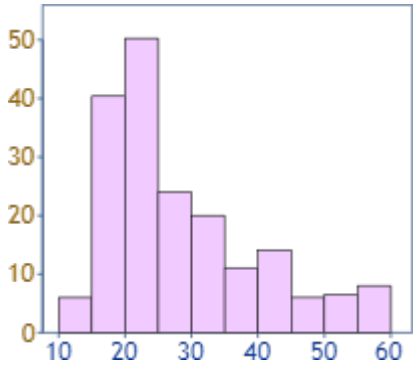
Normal Distribution

Data can be "distributed" (spread out) in different ways.

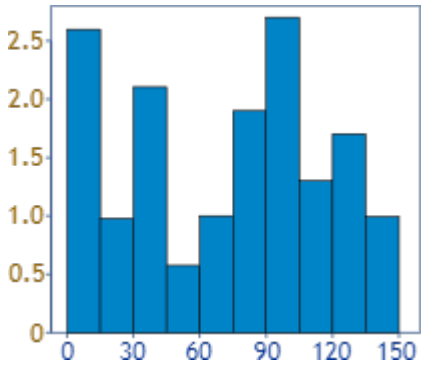
It can be spread out
more on the left



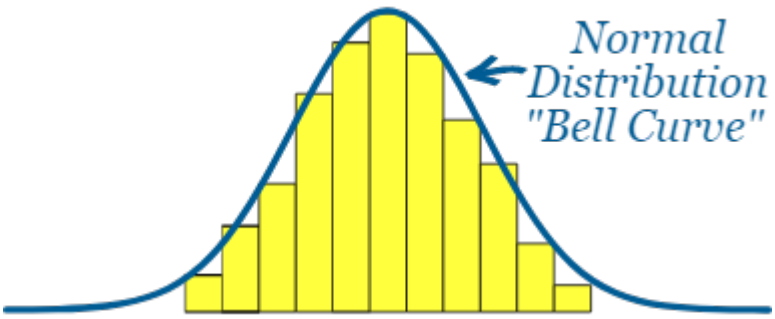
Or more on the right



Or it can be all jumbled up



But there are many cases where the data tends to be around a central value with no bias left or right, and it gets close to a "Normal Distribution" like this:



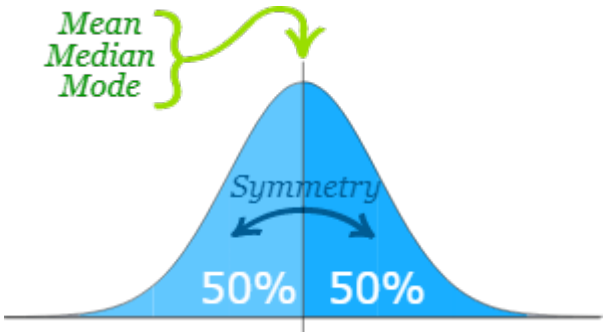
The blue curve is a Normal Distribution.
The yellow histogram shows some data that
follows it closely, but not perfectly (which is usual).

It is often called a "Bell Curve" because it looks like a bell.

Many things closely follow a Normal Distribution:

- heights of people
- size of things produced by machines
- errors in measurements
- blood pressure
- marks on a test

We say the data is "normally distributed":



The **Normal Distribution** has:

- mean = median = mode
- symmetry about the center
- 50% of values less than the mean and 50% greater than the mean

Chi Square Test

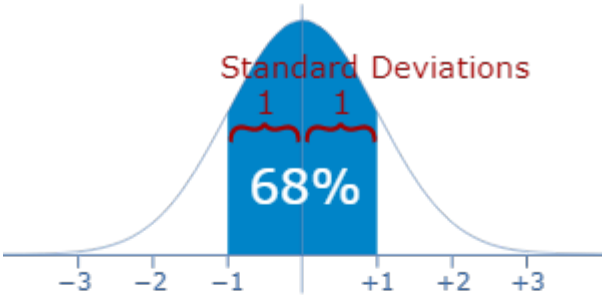
A **chi-squared test** (symbolically represented as χ^2) is basically a data analysis on the basis of observations of a random set of variables. Usually, it is a comparison of two statistical data sets. This test was introduced by **Karl Pearson** in 1900 for categorical data analysis and distribution. So it was mentioned as **Pearson's chi-squared test**.

The chi-square test is used to estimate how likely the observations that are made would be, by considering the assumption of the null hypothesis as true.

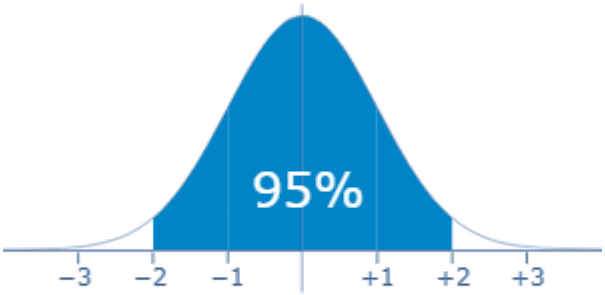
A hypothesis is a consideration that a given condition or statement might be true, which we can test afterwards. Chi-squared tests are usually created from a sum of squared falsities or errors over the sample variance.

Standard Deviations

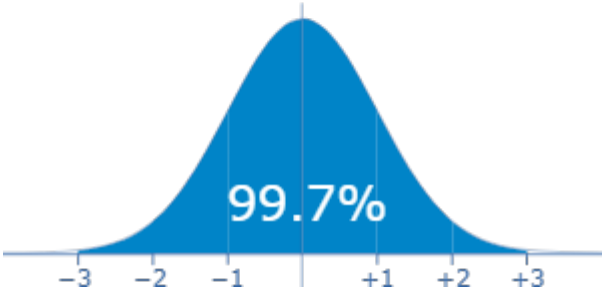
The Standard Deviation is a measure of how spread out numbers are (read that page for details on how to calculate it).When we calculate the standard deviation we find that **generally**:



68% of values are within
1 standard deviation of the mean

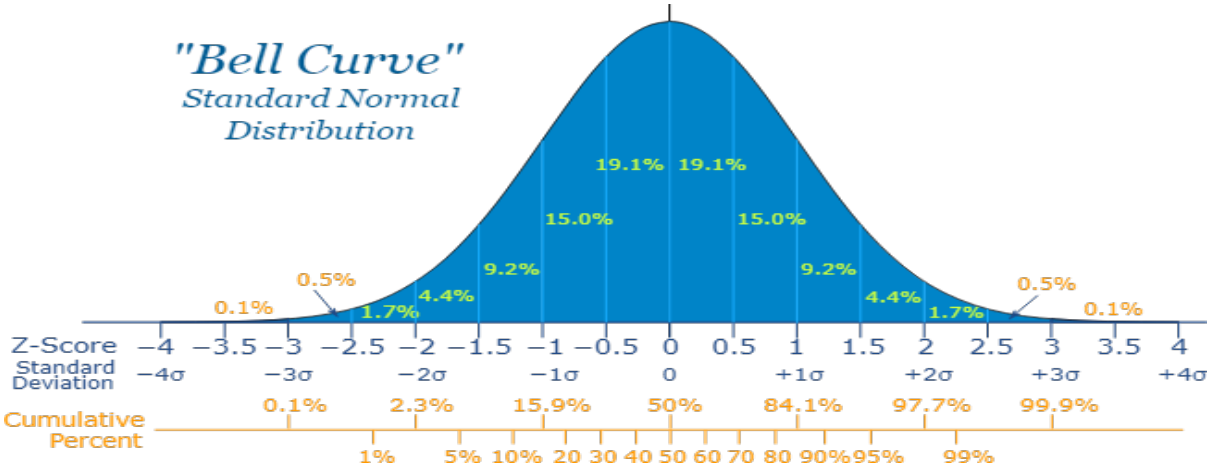


95% of values are within
2 standard deviations of the mean



99.7% of values are within
3 standard deviations of the mean

Here is the Standard Normal Distribution with percentages for every **half of a standard deviation**, and cumulative percentages:



New Balance - Company Introduction

New Balance Athletics, Inc. (NB), best known as simply New Balance, is one of the world's foremost sports footwear and apparel manufacturers. Based in Boston, Massachusetts, the multinational corporation was founded in 1906 as the New Balance Arch Support Company. New Balance maintains a manufacturing presence in the United States, as well as in the United Kingdom for the European market, where they produce some of their popular models. New Balance claims to differentiate its products with technical features, such as blended gel inserts, heel counters, and a greater selection of sizes, particularly for very narrow or extensive widths. The company is privately held and totaled $4.4 billion in revenue in 2021.

Data Source

The data source for heights of around 25000 people is taken from ‘kaggle’, an online community platform for data scientists and machine learning enthusiasts. This data has an index column and a height column.

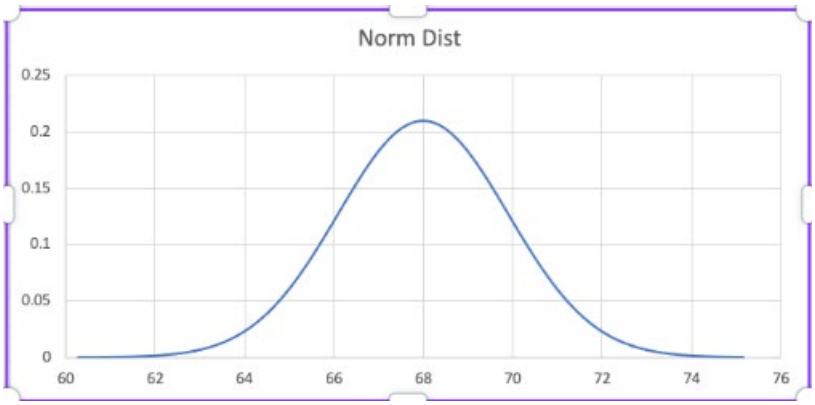
Challenges

New Balance wants to enter the Indian market and it does not have any relevant data regarding Indian male consumers' height dimensions to manufacture its products, quantity viz. The company wants to make sure that the size of its product fits an average Indian male customer.

Objectives

To obtain a suitable sample size to determine the average height of an Indian male customer proportionally. So that the decision can be taken by the manufacturers to manufacture the product such as hoodies, shirts, sweatshirts, trousers, shorts, etc. based on Indian male consumers.

Analysis



Mean - 68 inches

Std Dev. - 1.90 inches

Conclusion & Decision taken

- The average height as per the data set containing 25000 samples is 68 inches (5.6 ft).
- The standard deviation is 1.9
- The company decided to manufacture the male sportswear lineup according to Indian size.
- Where 34% of the total men's sportswear will be made for sizes for heights between 68-70 & 66-68 inches.
- 13.5 % of the total men's sportswear will be made for sizes, for heights between 70 - 72 & 64-66 inches.
- 2.35 % of the total men's sportswear will be made for sizes, for heights between 72-74 & 62-64 inches.
- 0.15 % of the total men's sportswear will be made for sizes, for heights between 74-76 & 60-62 inches.